INCH-POUND

MIL-DTL-17944B(SH)
2 July 2019

SUPERSEDING
MIL-C-17944A(SH)
27 January 1992

DETAIL SPECIFICATION

CAPSTANS, ELECTRIC - FOR SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric capstans for Naval shipboard use.

1.2 Classification. Capstans are of the following classes, as specified (see 6.2):

a. Class 1 – Capstan head mounted on weather deck with motor and gear reduction mounted on deck below.

b. Class 2 – Capstan head mounted on weather deck with motor and gear reduction mounted below deck in underhung position.

c. Class 3 – Capstan head, gear reduction, and motor mounted on weather deck.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

FF-S-85 - Screw, Cap, Slotted and Hexagon Head

FF-S-86 - Screw, Cap, Socket-Head

FF-S-92 - Screw, Machine, Slotted, Cross-Recessed or Hexagon Head

FF-S-200 - Setscrews: Hexagon Socket and Spline Socket, Headless

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3950

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FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52522 - Lining Brake, and Lining Material Brake: Automotive (Metric)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for
MIL-DTL-15024	-	Plates, Tags, and Bands for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification
MIL-DTL-16392	-	Brake, Electro-Mechanical, Naval Shipboard
MIL-DTL-17060	-	Motors, Alternating Current, Integral – Horsepower, Shipboard Use
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-PRF-23236	-	Coating Systems For Ship Structures
MIL-PRF-24635	-	Coating Systems, Weather-Resistant, Exterior Use
MIL-C-24707	-	Castings, Ferrous, General Specification for
MIL-DTL-24707/1	-	Castings, Ferrous, for Machinery and Structural Applications
MIL-C-24707/5	-	Castings, Ductile Iron and Austenitic Iron
MIL-DTL-32613	-	Controller, Auxiliary-System, Naval Shipboard

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1310	-	Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-860303 - Capstan and Gypsy Head, Tabulation of Sizes

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 20 - Radial Bearings of Ball, Cylindrical Roller and Spherical Roller Types, Metric Design

(Copies of this document are available online at www.americanbearings.org.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION

ANSI/AGMA 6001 - Design and Selection of Components for Enclosed Gear Drives

(Copies of this document are available online at www.agma.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ANSI/ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at www.asme.org.)

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ASTM INTERNATIONAL

ASTM A27/A27M	-	Standard Specification for Steel Castings, Carbon, for General Application
ASTM A131/A131M	-	Standard Specification for Structural Steel for Ships
ASTM A148/A148M	-	Standard Specification for Steel Castings, High Strength for Structural Purposes
ASTM A668/A668M	-	Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use
ASTM B505/B505M	-	Standard Specification for Copper Alloy Continuous Castings
ASTM F718	-	Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-STD-595	-	Colors Used in Government Procurement
SAE-AS35411	-	Fittings, Lubrication

(Copies of these documents are available online at www.sae.org.)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC-SP 10/NACE No. 2	-	Near-White Blast Cleaning

(Copies of this document are available online at www.sspc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.3 Material. Material shall conform to the following requirements as applicable:

a. Forged steel shall be in accordance with ASTM A668/A668M, grade D.

b. Cast steel shall be in accordance with MIL-C-24707, MIL-C-24707/1, ASTM A27/A27M, or ASTM A148/A148M.

c. Structural steel components shall be in accordance with MIL-S-22698, grade DH36 or ASTM A131/A131M.

d. Cast iron shall be in accordance with MIL-C-24707/5. Cast iron shall not be used for fabrication of the capstan head.

e. Copper alloy castings shall be in accordance with ASTM B505/B505M, alloy C86500 or C92300.

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3.4 Environmental conditions. Unless otherwise specified (see 6.2), electric capstans shall be operable and not lose lubrication or liquids on ships encountering ±15 degrees of static list, ±5 degrees of static trim, ±45 degrees of dynamic roll (combatants), ±30 degrees of dynamic roll (non-combatants), and ±10 degrees of dynamic pitch. Electric capstans shall be operable in temperatures ranging from -40 degrees Fahrenheit (°F) (-40 °C) to 140 °F (60 °C) with relative humidity from 0 to 100 percent.

3.4.1 Noise. Unless otherwise specified (see 6.2), electric capstan equipment shall not exceed MIL-STD-1474, grade D airborne sound level acceptance criteria.

3.4.2 Shock. Unless otherwise specified (see 6.2), all electric capstan system equipment shall meet MIL-DTL-901, grade B shock requirements.

3.4.3 Vibration. All rotating machinery shall be balanced in accordance with MIL-STD-167-1 to minimize vibration. Unless otherwise specified (see 6.2), the capstan shall operate continuously and reliably during and after exposure to type I vibration in accordance with MIL-STD-167-1 and shall withstand internally excited vibration in accordance with MIL-STD-167-1 type II vibration.

3.4.4 Electromagnetic interference (EMI) suppression. Electrical equipment and controllers shall comply with the EMI limits and criteria of MIL-STD-461 for below deck of surface ships for class 1 and 2 capstans, and for above deck of surface ships for class 3 capstans. Electrical equipment and controllers shall be bonded and grounded in accordance with MI-STD-461, MIL-STD-464, and MIL-STD-1310 to minimize adverse EMI effects and to protect personnel from electrical shock.

3.5 Construction. Each capstan shall consist of a capstan head, shaft, gear reduction, electric motor, controller, electric brake, and all operating accessories necessary for operation independent of anything except power leads (see 6.3).

3.5.1 Capstan arrangements. The capstan arrangement shall be of the class specified (see 1.2). It shall provide for clear leads of mooring and warping lines at all working angles to be handled by the capstan head. Unless otherwise specified (see 6.2), pawls are not required. Construction of the capstan shall eliminate hazards of exposed parts becoming inoperative due to frequent soaking by salt spray and extreme atmospheric conditions occurring on the weather deck of the vessel. Where capstan shafts pass through the deck, provisions shall be made to obtain thorough water tightness. Flexible couplings or slip joints shall be provided as necessary to compensate for deck deflections.

3.5.2 Welding. Welded fabrication properly made and stress relieved is acceptable in lieu of castings. Fabrication, welding, and inspection shall be in accordance with S9074-AR-GIB-010/278.

3.5.3 Weight. The unit shall be as compact and light as is consistent with strength and rigidity requirements. Full use shall be made of welded construction to obtain lightest weight. If required, maximum size dimensions and weight limits shall be as specified (see 6.2). In the absence of specific values, a compact and lightweight design efficiency compliant with the stress requirements of section 3.10 is preferred.

3.5.4 Fasteners. Positive means shall be provided for securing keys, feathers, splines, nuts, screws, and collars to prevent their working loose. The use of setscrews shall be avoided. When setscrews are used, they shall have positive locking devices to prevent their working out. Fasteners shall be in accordance with MIL-DTL-1222, FF-S-85, FF-S-86, FF-S-92, or FF-S-200. Use of ANSI standard fasteners meeting these cited specifications is permitted. Fasteners with a nominal diameter of 1.2 inch or less, and exposed with the unit in operating position, shall be of stainless-steel material. Screw threads shall be unified, coarse, or fine thread series in accordance with FED-STD-H28.

3.5.5 Capstan head. The capstan head shall be proportioned as per table I, figure 1, and the capstan head and rope size specified (see 6.2). The wearing surface shall be hardened to a minimum of 350 Brinell. Finished surface of the capstan head in contact with rope shall have surface roughness as defined in ANSI/ASME B46.1 of 125 microinches (3.2 micrometers) or less. The working area of the capstan head in contact with the line or rope shall be preserved to avoid corrosion but not painted.

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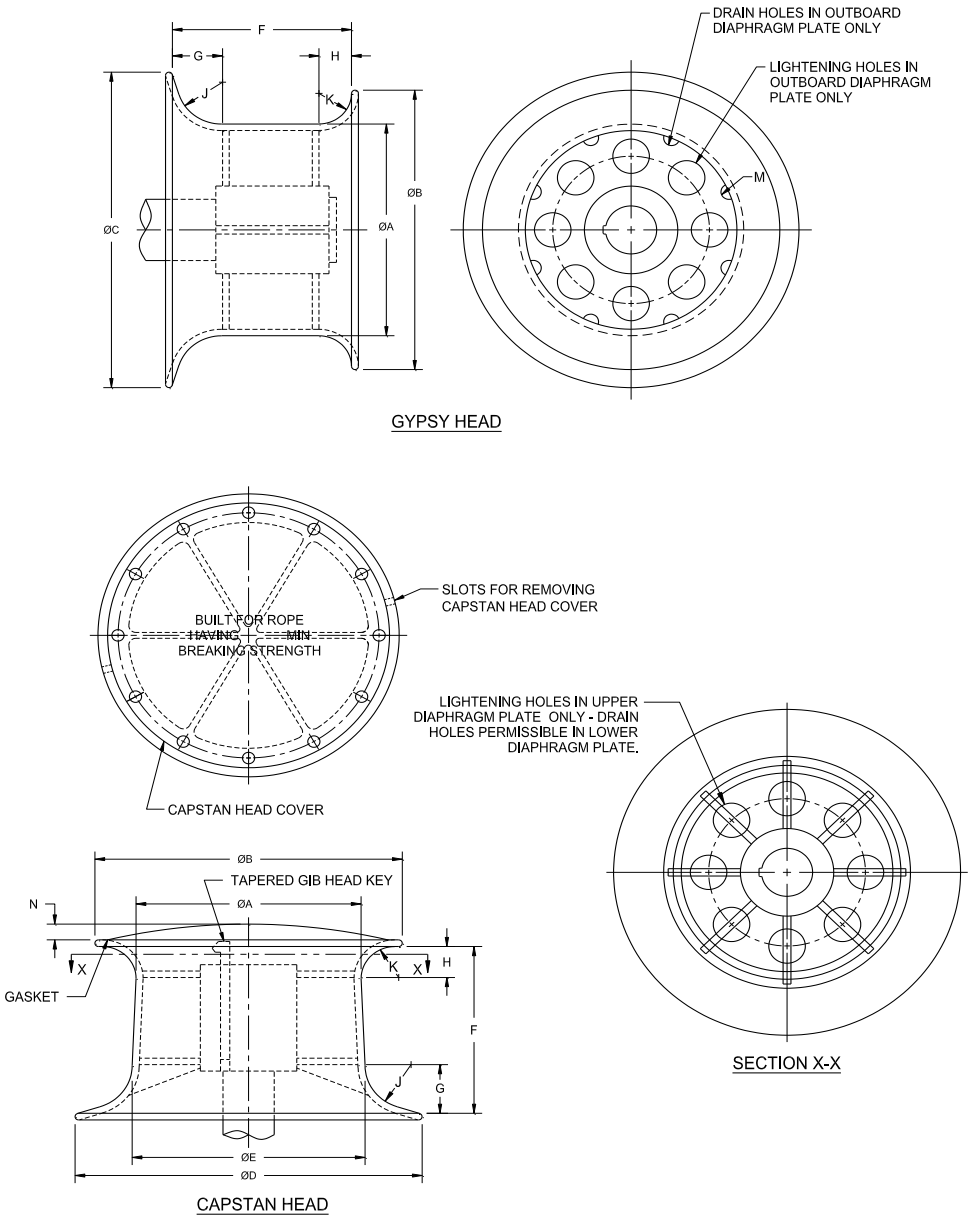


FIGURE 1. Capstan and gypsy head dimensional locations.

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TABLE I. Capstan and gypsy head dimensions.

Rope Minimum Breaking Strength (lbf) (*kN*)	**Capstan and Gypsy heads** Dimensions in Inches and *Millimeters* Per Figure 1												
	A dia	B dia	C dia	D dia	E dia	F	G	H	J rad	K rad	L approx.	M approx.	N
9000 *(40)*	9	12	13 5/8	13 5/16	9 3/16	7 3/16	1 7/8	1 1/4	1 5/8	1 1/8	7/16	5/8	3/8
	229	*305*	*346*	*354*	*233*	*183*	*48*	*32*	*41*	*29*	*11*	*17*	*10*
15000 *(67)*	12	16	18 3/16	18 5/8	12 1/4	9 5/8	2 1/2	1 11/16	2 3/16	1 1/2	5/8	13/16	13/32
	305	*406*	*462*	*473*	*311*	*244*	*64*	*43*	*56*	*38*	*16*	*21*	*10*
22500 *(100)*	15	20	22 3/4	23 1/4	15 5/16	12	3 1/8	2 1/8	2 3/4	1 7/8	3/4	1	1/2
	381	*508*	*578*	*591*	*389*	*305*	*79*	*54*	*70*	*48*	*19*	*25*	*13*
31000 *(138)*	18	24	27 5/16	27 3/8	18 3/8	14 3/8	3 3/4	2 9/16	3 5/16	2 1/4	7/8	1 3/16	39/64
	457	*610*	*694*	*695*	*47*	*365*	*95*	*65*	*84*	*57*	*22*	*30*	*15*
41000 *(182)*	21	28	31 7/8	32 9/16	21 3/8	17	4 3/8	3	3 7/8	2 5/8	1 1/16	1 3/8	45/64
	533	*711*	*810*	*827*	*543*	*432*	*111*	*76*	*98*	*67*	*30*	*35*	*18*
52000 *(231)*	24	32	36 3/8	37 3/16	24 7/15	19 3/16	5	3 3/8	3 7/8	2 5/8	1 1/16	1 5/8	51/64
	610	*813*	*924*	*945*	*621*	*487*	*127*	*86*	*98*	*67*	*27*	*41*	*20*
64000 *(285)*	27	36	40 5/16	41 7/8	27 1/2	21 5/8	5 5/8	3 3/16	4 15/16	3 3/8	1 3/8	1 13/16	29/32
	686	*914*	*1040*	*1064*	*699*	*549*	*143*	*97*	*125*	*86*	*35*	*46*	*23*
77000 *(343)*	30	44	45 1/2	46 1/2	30 9/16	24	6 1/4	4 1/4	5 1/2	3 3/4	1 1/2	2	1
	762	*1016*	*1156*	*1181*	*776*	*610*	*159*	*108*	*140*	*95*	*38*	*51*	*25*
91000 *(405)*	33	44	50 1/16	51 1/8	33 5/8	26 3/8	6 7/8	4 11/16	6 1/16	4 1/8	1 5/8	2 3/16	1 3/32
	838	*1118*	*1272*	*1299*	*854*	*670*	*175*	*119*	*154*	*105*	*41*	*56*	*28*
120000 *(534)*	36	48	54 5/8	55 3/16	36 11/16	28 13/16	7 1/2	5 1/8	6 5/8	4 1/2	1 13/16	2 3/8	1 7/32
	914	*1219*	*1387*	*1418*	*932*	*732*	*191*	*130*	*168*	*114*	*46*	*60*	*31*

NOTE:

1. For further detail see 803-860303.

3.5.6 Gearing. Gearing shall be in accordance with ANSI/AGMA 6001, shall have machine cut teeth, and shall be provided with guards in accordance with MIL-STD-1472 for the protection of personnel and the gears. Gears shall be totally enclosed in an oil-tight case and shall be lubricated by an oil bath. Oil fill, drain, and vent fitting shall be provided. Housings containing gears shall be provided with oil-tight covered openings in sufficient numbers to allow visual examination of the teeth on all gears. Such openings shall expose at least two gear teeth at a time. The opening shall permit viewing the mesh between gears, if practical. A dipstick, sightglass, or plug shall be provided for indicating the oil level. Dipsticks shall be marked to indicate “full” and “add oil.”

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3.5.7 Bearings. Where practicable, anti-friction bearings in accordance with ABMA 20, shall be used throughout. However, if it is not practicable, bronze bushings shall be used. Where bronze bushings are used, positive means shall be provided to prevent the bushing from turning and cutting off the supply of lubricant. Each bearing, except those lubricated by an oil bath, shall be fitted with pressure type grease fittings in accordance with SAE-AS35411.

3.5.8 Bedplates. The bedplates shall be capable of maintaining alignment without aid from the deck foundations. The hold-down bolt spacing shall not exceed 15 times the bolt diameter, and the thickness of the flange at these bolt holes shall be drilled 1/16 inch (1.6 millimeter) small for reaming out at installation to fit body bound bolts. The configuration shall eliminate pockets where water can stand or collect.

3.5.9 Accessibility. Capstan shafts, gears, bearings, and heads shall be constructed to permit inspection, adjustment, or removal without dismantling the entire unit.

3.5.10 Motors. Motors and motor characteristics shall be in accordance with MIL-DTL-17060 or as specified (see 6.2). The motor load rating shall be selected so that the 100 percent rating shall not be exceeded at the capstan's rated load and speed. Unless otherwise specified (see 6.2), alternating current (AC) electric capstans shall be powered by two-speed AC motors rated at full and one-half speeds and constant horsepower.

3.5.11 Motor controllers. Controllers and master switches shall be in accordance with MIL-DTL-32613, including cybersecurity requirements, or as specified (see 6.2). An emergency run feature shall be provided in the control group. Master switches shall be spring-returned to "off." Master switches shall also be deck mounted as near as practicable to the capstan they control without jeopardizing the safety of the operator from synthetic line snapbacks.

3.5.12 Electric brakes. Brakes shall be in accordance with MIL-DTL-16392 and as specified (see 6.2). Brakes shall be able to stop and hold within 3 feet (1 meter) of the capstan at 150 percent rated line pull. Brake linings shall be made of a material in accordance with A-A-52522.

3.6 Painting and preservation procedure. Painting and preservation procedure. Unless otherwise specified herein or on manufacturers NAVSEA reviewed ASTM F718 data sheets, machinery and equipment shall be painted and preserved at ambient conditions as follows: air and surface temperature between 50 and 100 °F (10 and 38 °C), and relative humidity not greater than 85 percent. Coatings shall be applied only when the ambient temperature and the temperature of the prepared substrate are 50 °F (10 °C) or greater and the temperature of the prepared substrate is a minimum of 5 °F (-15 °C) above the dew point. Coatings shall be applied to equipment shortly after being manufactured and cleaned. Metal to be coated shall have surfaces abrasive blasted to a "near white metal" level of cleanliness in accordance with SSPC-SP 10/NACE No. 2. The abrasive blast surface profile shall be 2 to 4 mils. Coatings shall be cured to overcoating and to service in accordance with the NAVSEA-reviewed ASTM F718 data sheets.

3.6.1 Painted surfaces. Capstan external surfaces shall be primed with two coats of epoxy in accordance with MIL-PRF-23236, type V/VI, class 5 or 7 at 4 to 8 mils dry film thickness (DFT) per coat and one coat of polysiloxane topcoat in accordance with MIL-PRF-24635, type V/VI, class 2, grade B or C at 5 to 8 mils DFT per coat. The MIL-PRF-24635 coating shall be SAE-AMS-STD-595, color number 26270, haze gray. Unless otherwise specified (see 6.2), coatings shall be applied in accordance with the coating manufacturer's NAVSEA-reviewed ASTM F718 product data sheet.

3.6.2 Unpainted surfaces. Bearing and machined surfaces, interior "oil-wetted" surfaces, and corrosion-resistant material shall not be painted.

3.7 Capstan performance. Capstan performance requirements shall be as specified herein or in the contract (see 6.2). Capstans shall be of the reversible type and shall be operable and shall develop specified lines pulls and speeds in both directions of rotation. The installed capstan head, shafting, bearings, gears, and bedplate shall withstand a static load equal to the minimum line-breaking strength, mid-height and normal to its axis. Each part of the capstan shall be able to withstand the greater of either a torque causing slippage of the brake or stalling of the driving machinery. This torque and the static load shall not be considered to act simultaneously.

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3.8 Identification plates and marking. Identification and marking shall be in accordance with MIL-STD-130. Identification plates and other designated marking plates shall be in accordance with MIL-DTL-15024 and MIL-P-15024/5. Plates, including operating and lubrication charts exposed to the weather, shall be fabricated from corrosion-resistant steel, and shall be type A, B, or C. Charts shall be mounted by being fully bedded with adhesive and attached with corrosion-resistant steel fasteners.

3.9 Special tools. When specified (see 6.2), a set of special tools that is necessary for the proper maintenance of the equipment shall be included. Each special tool shall be marked for its intended service. Special tool markings shall be in accordance with MIL-STD-130. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

3.10 Allowable stresses. The strength relationships specified in table II shall be used in design stress calculations.

TABLE II. Allowable stress.

Design strength	Percent of the tensile yield point
Direct shear	60
Torsional shear	60
Compression (bearing)	160

3.10.1 Rated load condition stress. When subjected to the rated load, the combined stresses in any part of the capstan shall not exceed 35 percent of the tensile yield point of the ductile materials in any normal operating condition (see 6.2). Rated conditions for these calculations for ac multiple speed drives shall be those for low speed operation.

3.10.2 Breaking strength load stress. Stresses shall not exceed 70 percent of the tensile yield point in any part of the equipment when the capstan head is subjected to a horizontal pull equal to the minimum breaking strength of the synthetic rope as specified (see 6.2). Additionally, stress shall not exceed 70 percent of the tensile yield when the synthetic rope is acting at mid-height of the capstan head and normal to the axis of the shaft.

3.10.3 Motor torque load stress. When the capstan head is subjected to a pull equal to the stalling torque of the motor, the combined stresses shall not exceed 70 percent of the tensile yield point in any part of the machine.

3.11 Reliability and maintainability. Units shall meet or exceed the following requirements or be as specified (see 6.2):

a. Mean time between failure: 10,000 operational uses. An operational use is defined as 10 minutes at rated load.

b. Unit design life: 30 years.

c. Mean time to repair: 1.0 hours.

d. Minimum time between maintenance: 250 operational uses.

e. Statistical confidence: 80 percent.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on electric capstans when a first article sample is required (see 3.1). This inspection shall include the tests as specified in table III.

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4.3 Conformance inspection. Conformance inspection shall include the tests as specified in table III.

TABLE III. First article and conformance inspection and testing.

Characteristic	Requirement	First article	Conformance
Materials	3.3	4.5	---
High temperature	3.4	4.6.1.5	---
Low temperature	3.4	4.6.1.6	---
Humidity	3.4	4.6.1.7	---
Noise	3.4.1	4.6.1.1	---
Shock	3.4.2	4.6.1.2	---
Vibration	3.4.3	4.6.1.3	---
EMI	3.4.4	4.6.1.4	---
Construction, painting, identification plates and special tools	3.5, 3.7, 3.8, and 3.9	4.5	4.5
Performance	3.6	4.6.3.1.a, and 4.6.3.1.b	4.6.3.1.a
Stresses	3.10	4.6.3.1.b	---
Reliability and maintainability	3.11	4.6.2.1 and 4.6.2.2	---

4.4 Test conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6. The equipment is not required to operate under full-load conditions for more than 1 hour in a 4-hour period. Unless otherwise specified (see 6.2), environmental conditions shall meet MIL-STD-810 Coastal/Ocean, Exposed (for weather deck machinery), and Sheltered (for interior machinery) criteria.

4.5 Examination. Electric capstans shall be visually examined for compliance with the requirements specified in 3.5 through 3.5.12, 3.7, 3.8, and 3.9. This inspection shall encompass all visual examinations and dimensional measurements.

4.6 Tests. Each capstan shall be tested as noted in table III to verify conformance to the requirements of this specification.

4.6.1 Environmental tests. Unless otherwise specified (see 6.2), environmental tests of 4.6.1.1 through 4.6.1.7 shall be conducted during first article testing.

4.6.1.1 Airborne noise test. The airborne noise shall be measured in accordance with the instrumentation and procedures of MIL-STD-1474 with the exception that noise levels at the measurement positions shall be averaged. Motors shall be tested while operating at no load with rated voltage and frequency applied.

4.6.1.2 Shock test. Components shall be tested for high-impact shock in accordance with MIL-DTL-901.

4.6.1.3 Vibration test. The electric capstan shall be tested in accordance with MIL-STD-167-1 type I, ranging from 4 hertz (or lowest attainable frequency) through 50 hertz, and type 2.

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4.6.1.4 EMI test. The emission and susceptibility tests specified in MIL-STD-461 shall be performed to determine electromagnetic compliance with 3.4.4.

4.6.1.5 High temperature test. The equipment shall be subjected to a high-temperature test in accordance with MIL-STD-810 method 501, procedure II. The cyclic temperature test shall be performed.

4.6.1.6 Low temperature test. The equipment shall be subjected to a low-temperature test in accordance with MIL-STD-810 method 502, procedure I.

4.6.1.7 Humidity test. The equipment shall be subjected to a humidity test in accordance with MIL-STD-810 method 507.

4.6.2 Reliability and maintainability tests.

4.6.2.1 Reliability. The reliability of the capstan shall be demonstrated by the successful completion of all operational tests (see 4.6.3) and reliability predictions. The guidance provided in MIL-HDBK-217 may be used.

4.6.2.2 Maintainability. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.6.3 Operational tests.

4.6.3.1 Shop tests.

a. No load. Each unit shall be run under no load for a period of 30 minutes at each speed in each direction and observed to determine that no abnormal heating, wear, or noise shall occur. No part of the equipment shall show any sign of permanent distortion or undue wear after this test.

b. Rated load. One unit shall be selected from each lot offered for delivery and tested at rated load and speeds for one-half hour in both directions of rotation. The electric brake shall be tested by stopping and holding within 3 feet of the capstan with 150 percent rated line pull. No part of the equipment shall show any sign of permanent distortion or undue wear after this test. Compliance with the requirements of 3.10 shall be demonstrated by stress gauge instrumentation or by calculation.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The electric capstan is intended for use in handling lines when mooring, warping, towing, boat handling and other related uses aboard naval ships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Class required (see 1.2).

c. When first article is required (see 3.1).

d. Environmental conditions, if different from specified herein (see 3.4, 3.4.1, 3.4.2, and 3.4.3).

e. If pawls are required (see 3.5.1).

f. Maximum size and weight if required (see 3.5.3).

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g. Capstan head and rope size (see 3.5.5).
h. Characteristics of motors (see 3.5.10).
i. If AC electric capstans are to be powered other than as specified (see 3.5.10).
j. Characteristics of controllers (see 3.5.11).
k. Characteristics of brakes (see 3.5.12).
l. If surfaces are to be painted other than as specified (see 3.6.1).
m. Performance requirements (see 3.7 and 3.10.2).
    (1) Rated load and line speed.
    (2) Maximum load at no specified speed (breaking strength of intended mooring line).
    (3) Static loaded at mid-height of capstan head.
n. Special tools, if required (see 3.9).
o. Allowable stresses if different from herein (see 3.10.1).
p. Reliability and maintainability requirements if different from herein (see 3.11).
q. If enviormental test conditions are other than as specified (see 4.4).
r. If environmental tests are other than as specified (see 4.6.1).
s. Packaging requirements (see 5.1).
t. Technical manuals (see 6.3).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied in a contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 Subject term (key word) listing.

Allowable stress
Brake, electric
Capstan head
Gear reduction

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 3950-2019-001)

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